Involution blocks and generation in the small Ree groups

Human Compression project: a partial recovery of L039

This note supplies complete proofs for the rank-three construction and the odd-quotient exclusion in Leemans–Schulte–Van Maldeghem [1]. A centralizer criterion also recovers their upper bound of four, using an explicitly retained structural result. The rank-four exclusion remains unrecovered here. We isolate a difficulty in its printed proof; neither that difficulty nor the results below refute the paper's principal theorem.

1 Three elementary criteria

Lemma 1 (Odd quotients)**.** *If $N \trianglelefteq G$ and G/N has odd order, every involution of G belongs to N. In particular, G cannot be generated by involutions if $N \neq G$.*

Proof. The image of an involution has order dividing two and the odd number $|G/N|$, hence is the identity. Every product of involutions is therefore in N. □

Lemma 2 (Centralizer intersections)**.** *Let G be a nonabelian simple group. Suppose that $C_G(a) \cap C_G(b)$ is abelian whenever a, b are noncommuting involutions. Then no generating string of involutions for G has length at least five.*

Proof. Suppose $(r_0, \ldots, r_{d-1})$ is such a string with $d \geq 5$. If r_0 commuted with r_1, it would commute with every generator, contradicting the trivial center of G. Similarly r_{d-2} and r_{d-1} do not commute. String commutation puts both of the latter in $C_G(r_0) \cap C_G(r_1)$, an abelian group, a contradiction. This argument needs no intersection property. □

Lemma 3 (Coprime adjacent orders)**.** *Let $G = \langle a, b, c\rangle$, where a, b, c are distinct involutions, $ac = ca$, and $|ab| = m$, $|bc| = n$ are coprime odd integers greater than one. Suppose G is larger than each of its three two-generator subgroups. Then (a, b, c) has the intersection property.*

Proof. The adjacent subgroups are dihedral of orders $2m$ and $2n$. Their intersection contains $\langle b\rangle$ and has order dividing $2\gcd(m, n) = 2$, so equals $\langle b\rangle$. The third parabolic is $\langle a, c\rangle \cong C_2^2$. Its intersections with the adjacent parabolics have order dividing $\gcd(4, 2m) = \gcd(4, 2n) = 2$ and contain the prescribed common endpoint. An omitted singleton generator cannot lie in its opposite parabolic, since then that proper subgroup would contain every generator. Distinct singleton subgroups intersect trivially. Together with the containment, empty-set and whole-group cases, these observations give the assertion for every pair of generator subsets. □

2 A partition by fixed blocks

Let the finite set Ω carry a Steiner 2-design whose blocks all have the same even size k. Suppose a group acts on this design and its involutions are in bijection with the blocks, an involution u corresponding to its full fixed set B_u. Thus distinct involutions have at most one common fixed point.

Lemma 4. *Fix an involution t. The blocks of the involutions in $C_G(t)$ are pairwise disjoint. If that centralizer has exactly $|\Omega|/k$ involutions, those blocks partition Ω.*

Proof. For $u \in C_G(t) \setminus \{t\}$, the set B_t is invariant under u. The number of its points fixed by u is even, since k is even and all nonfixed orbits have size two. That number is at most one, so is zero. Hence $B_t \cap B_u = \varnothing$.

Now take distinct $u, v \in C_G(t) \setminus \{t\}$. If p belonged to $B_u \cap B_v$, then $p \notin B_t$, so $p^t \neq p$. Commutation makes p^t another common fixed point of u and v, contrary to the Steiner condition. Thus all the blocks are disjoint. Under the numerical hypothesis, their union has $(|\Omega|/k)k = |\Omega|$ points. □

The small Ree group $G = R(q)$, $q = 3^{2e+1} \geq 27$, acts on its unital, a Steiner 2-design with $|\Omega| = q^3 + 1$ and $k = q + 1$. Its involution centralizer is $\langle t \rangle \times \mathrm{PSL}_2(q)$. We retain these classical structural facts, including the block–involution bijection. The usual involution class calculation in $\mathrm{PSL}_2(q)$, where $q \equiv 3 \pmod 4$, gives

$$\frac{|\mathrm{PSL}_2(q)|}{q+1} = \frac{q(q-1)}{2}$$

involutions. Consequently $C_G(t)$ has $1 + q(q-1)$ involutions, counting t and the two lifts of each involution in the second factor. Since

$$(q^2 - q + 1)(q + 1) = q^3 + 1,$$

Lemma 4 applies. In particular each point outside B_t is fixed by exactly one involution different from t that commutes with t. The partition conclusion already appears in Pierro [2, Lemma 2.2 and the following paragraph]; the proof above records the direct fixed-point argument. It is not a new result claimed by this project.

3 The rank-three marking

We retain the standard subgroup list for $R(q)$ used in Section 2.2 of [1]. Put

$$\alpha = q + 1 - \sqrt{3q}, \qquad \beta = q + 1 + \sqrt{3q}.$$

The two nonsplit cyclic Hall subgroups have orders α, β and normalizers of the form $C_\alpha : C_6$ and $C_\beta : C_6$. The other maximal subgroups are point stabilizers $U : C_{q-1}$, involution centralizers $C_2 \times \mathrm{PSL}_2(q)$, four-group normalizers of order $6(q+1)$, and proper subfield Ree groups. We also retain the element-order list: besides $2, 3, 6, 9$, nonidentity orders divide $q - 1$, $(q + 1)/2$, α, or β. These facts are summarized independently in [2, Section 2, Tables 1–3].

Proposition 5. *For every $q = 3^{2e+1} \geq 27$, the group $R(q)$ has a string C-group marking (t, u, v) with $|tu| = \beta$ and $|uv| \in \{3, 9\}$.*

Proof. Let $T = \langle x \rangle$ have order β and let $M = N_G(T) = T : C_6$. The involution of its complement inverts T. Thus its involutions generate a dihedral subgroup D of order 2β. Choose involutions $t, u \in D$ with $tu = x$.

We first verify that M is the unique maximal subgroup containing x. We have

$$\alpha\beta = q^2 - q + 1, \quad \gcd(\alpha, \beta) = \gcd(\beta, q - 1) = \gcd(\beta, q + 1) = \gcd(\beta, 6) = 1.$$

For the first coprimality, a common divisor divides $2\sqrt{3q}$ and is prime to six, so is one. For the other two, reduce $q^2 - q + 1$ modulo $q - 1$ and $q + 1$; the remainders are respectively 1 and 3. Hence all the other listed maximal-subgroup types except the subfield groups cannot contain an element of order β. A proper subfield has parameter q_0 with $q \geq q_0^3$. Every element order in $R(q_0)$ is at most $\max(9, q_0 + \sqrt{3q_0} + 1)$, which is smaller than β; this also covers $q_0 = 3$. Finally, in any normalizer $T' : C_6$ with $|T'| = \beta$, the image of x in the quotient is trivial, since its order is prime to six. Thus $T' = \langle x \rangle = T$, and that normalizer is M.

The blocks B_t and B_u are disjoint. Otherwise D fixes their common point, whereas a point stabilizer of order $q^3(q-1)$ has no element of order β. Choose $p \in B_u$. The partition lemma gives a unique involution $v \neq t$ in $C_G(t)$ fixing p. It also gives $v \neq u$, since t and u do not commute. The product uv is a nonidentity element in the normal 3-subgroup of G_p: both involutions have the same nontrivial image in its cyclic quotient C_{q-1}. That 3-subgroup has exponent nine by the retained order list, so $|uv| = 3$ or 9.

Every involution in $M = T : C_6$ lies in the dihedral preimage of the unique order-two subgroup of C_6. The product of any two such involutions lies in T and has order dividing β. Since $|uv| \in \{3, 9\}$, it follows that $v \notin M$. The unique-maximal property of $x = tu$ now proves $\langle t, u, v\rangle = G$. The adjacent orders are coprime odd numbers and t commutes with v. Simplicity excludes equality of G with any of the dihedral or Klein-four parabolics. Lemma 3 proves all intersections. □

This recovers source Lemma 15 with its missing choice of v justified. In its last generation sentence the printed $\rho_0 \notin M$ cannot hold, since ρ_0 was chosen in M; the argument requires $\rho_2 \notin M$, which has been proved above.

4 The upper bound and the remaining obstruction

The classical description $\mathrm{Aut}(R(q))/R(q) \cong C_{2e+1}$ shows by the odd-quotient lemma that no proper almost simple extension is generated by involutions. Likewise $R(3) \cong \mathrm{PSL}_2(8) : C_3$ is not generated by involutions. This recovers source Lemma 2 and the stated nonsimple exception.

For completeness about the rank bound, we retain Pierro's structural intersection result [2, Lemma 3.13, noncommuting case]: for noncommuting involutions a, b of a simple small Ree group, $C_G(a) \cap C_G(b)$ is elementary abelian of characteristic three, or is a subgroup of a Klein four-group. It is therefore abelian. Lemma 2 immediately excludes ranks at least five, recovering source Lemma 3. The structural intersection theorem remains an external input whose cost must be included in any subsequent measurement.

The rank-four exclusion in source Lemmas 9–14 remains a separate task. There is a precise issue in the last paragraph of the proof of Lemma 14. It identifies

$$N = N_G(D) = (E_4 \times D) : C_3, \qquad H = D : C_3,$$

and says that H is normal in N. Under the stated four-group action, the order-three factor permutes the three nonidentity elements of E_4; see also [2, Section 2.2.4]. Since $D \trianglelefteq N$, this gives

$$N/D \cong A_4, \qquad H/D \cong C_3.$$

The latter is not normal in A_4: its four order-three subgroups are conjugate and distinct. Normality passes to quotients, so the displayed normality assertion cannot hold with these simultaneous identifications. In particular its use to force the same H into two overgroups needs a different argument. This elementary obstruction does not prove that rank-four Ree polytopes exist, and we make no such claim.

The present note proves only the stated rank-three, outer-extension, nonsimple and rank-at-most-four conclusions, with all borrowed inputs identified. The full principal-result contract of L039 is still pending. There is no new conjecture answer, Lean formalization or compression factor.

References

[1] D. Leemans, E. Schulte and H. Van Maldeghem, *Groups of Ree type in characteristic 3 acting on polytopes*, retained arXiv:1501.01125v1, eighteen pages, body dated 10 May 2018.

[2] E. Pierro, *The Möbius function of the small Ree groups*, Australas. J. Combin. **66**(2) (2016), 142–176, `https://ajc.maths.uq.edu.au/pdf/66/ajc_v66_p142.pdf`.